08. ORIGIN OF THE RETINA AND THE LENS

DURATION : 04:24

STUDY SHEET

COURSE SUMMARY

Examination of ocular embryology, from retinal evolution to the lens. Highlighting juxtacrine communications and interactions between the epiblast, CSF, and amniotic fluid. Study of the primary and secondary optic vesicle, and the transformation of hyaloid vessels into the retinal network (brain extension modulated by chemical gradients). Lens organogenesis is linked to the surface epithelium and the partitioning of ocular chambers.

ORIGIN OF THE RETINA AND LENS

The evolution of the eye begins with the observation of a **space** between the different structures. The **epiblast** attaches and changes shape, while inside, **cerebrospinal fluid (CSF)** and **amniotic fluid** are found, which have a common origin.

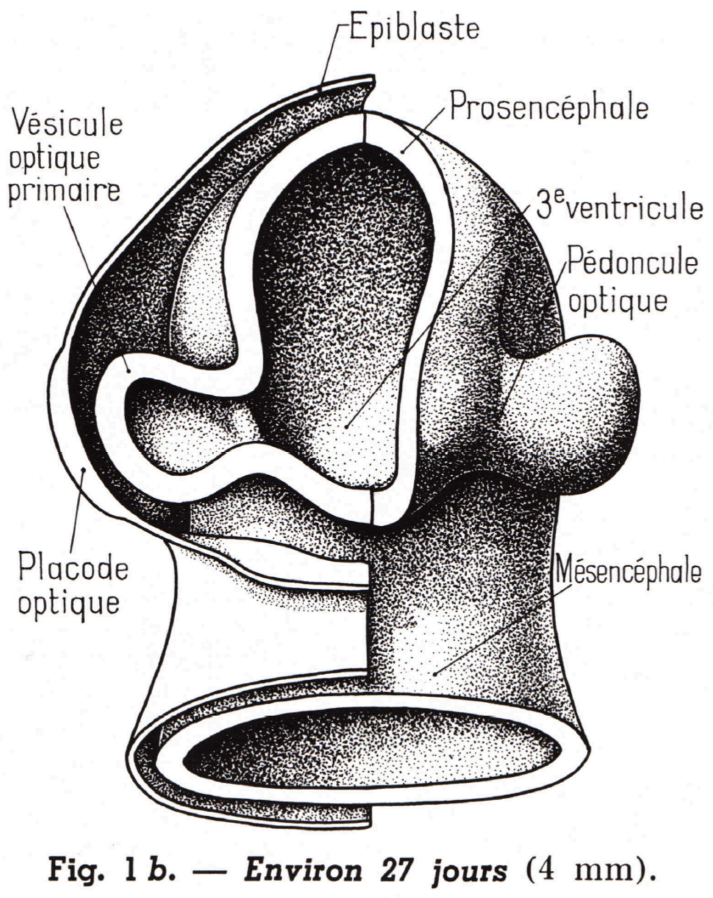

FIG. — First stages of eye development

Juxtacrine communication is established, allowing for molecular connections. The path of a vessel is also observed, representing the first wave of the **choroid**, the vascular tissue of the eye. This invasion is crucial for the formation of an artery, involving various components such as the presence of **vesicles** and a **chemical gradient**.

The choroidal fissure allows the passage of **hyaloid vessels**, which will later transform into **retinal vessels**. The communication between the epiblast and this area gives rise to the **primary optic vesicle**, which evolves into the **secondary optic vesicle**. This tissue divides into the **inner layer** and **outer layer** of the retina, thus forming the **retinal space**.

Anatomists and embryologists consider the retina to be an extension of the **brain**, influenced by cerebrospinal fluid and information exuded on an intercellular plane. The **lens vesicle**, which will form the **lens**, also originates from the surface epiblast. This encounter occurs through juxtacrine communication.

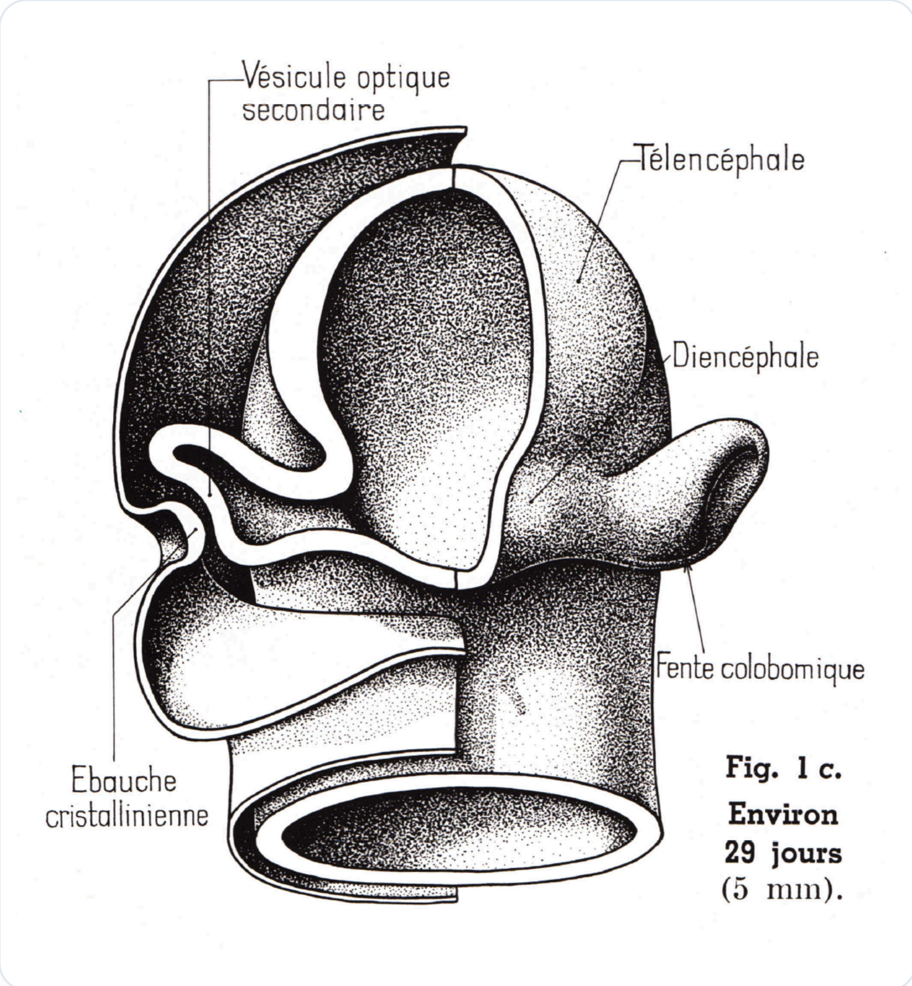

FIG. — Formation of the optic vesicle and retinal differentiation

The lens forms internally, while the surface epithelium remains in place, following a pattern similar to that of **neural tube** formation. The skin, on the surface, divides into different layers. The lens tissue constitutes the first layer of the eye.

The **conjunctiva** is the first membrane that covers the eye, also lining the future eyelids. Inside the lens, a space divides into two chambers: **anterior** and **posterior**. The retina, meanwhile, is extremely thin, similar to cigarette paper.

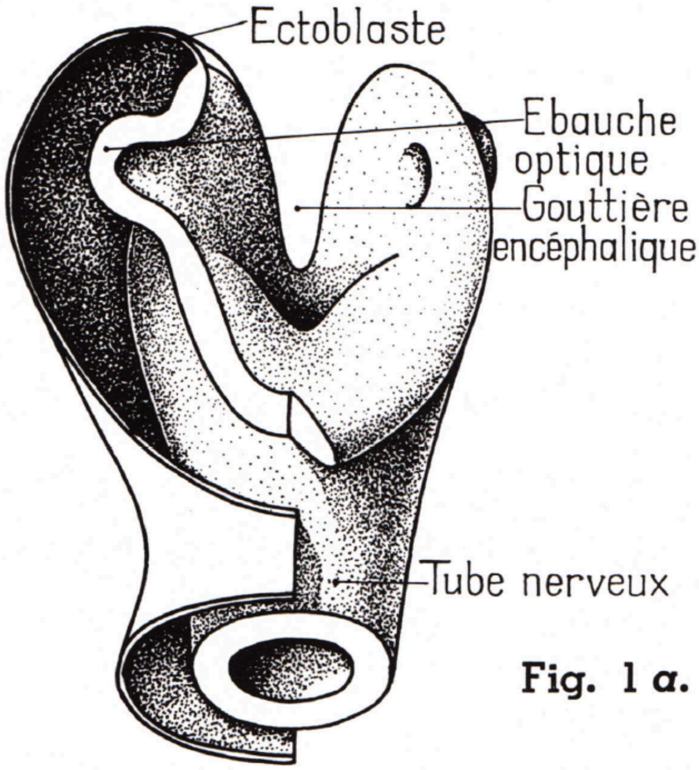

Fig. 1 a.

Environ 21 jours (2 mm).
L'embryon est vu de face.

FIG. — *Lens development and invagination*